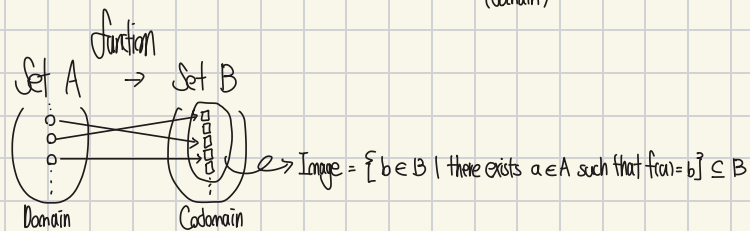


< Mathematical Functions >

< Notes on functions >

A function is a way of associating an element of the source set to exactly one element of the target set.

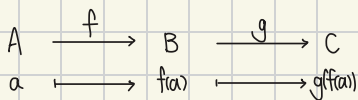


Injectivity & Surjectivity

* A function $f: A \rightarrow B$ is called injective if $f(a) = f(b) \rightarrow a = b$
 surjective if $\forall b \in B \exists a \in A$ that $f(a) = b$
 bijective if it's both injective and surjective.

Composition

* Given $f: A \rightarrow B$ and $g: B \rightarrow C$, we can construct the composition $(g \circ f): A \rightarrow C$ defined by $(g \circ f)(a) = g(f(a)) \in C$.



$\cdot g \circ f$ is injective $\Leftrightarrow f$ is injective & $g|_{\text{Im}(f)}$ is injective

$\cdot g \circ f$ is surjective $\Leftrightarrow g|_{\text{Im}(f)}$ is surjective

Inverse

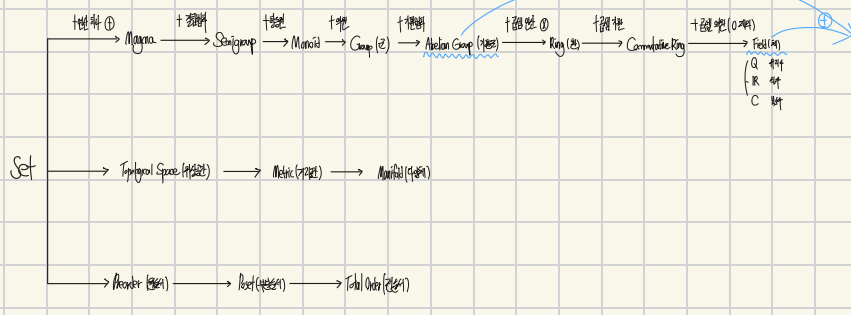
* Let $f: A \rightarrow B, g: B \rightarrow C$.

If $g \circ f = \text{Id}$, we say that g is a left inverse of f , and f is a right inverse of g .

- $\cdot f$ has a left inverse ($\exists g$ such that $g \circ f = \text{Id}$) $\Leftrightarrow f$ is injective (since Id is bijective)
- $\cdot f$ has a right inverse ($\exists g$ such that $f \circ g = \text{Id}$) $\Leftrightarrow f$ is surjective

If $g \circ f = \text{Id}$ and $f \circ g = \text{Id}$, f is an inverse of g , and g is an inverse of f .

- $\cdot$ If the function f has an inverse, that inverse is unique (Uniqueness of inverses: ~~if there is more than one, they are all the same~~)
- $\cdot$ If a function f has an inverse, f is bijective.



"Vector Space" = 일정한 구조를 가진, 벡터 연산을 할 수 있는 집합 $\Rightarrow$ Vector = Anything when there's a notion of adding two vectors & multiplying two vectors by a number.

- V is $\mathbb{R}$ or $\mathbb{C}$ or $\mathbb{H}$ or any other field
- Commutative: $u \cdot v = v \cdot u$
- Associative: $(u \cdot v) \cdot w = u \cdot (v \cdot w)$
- Distributive: $u \cdot (v + w) = u \cdot v + u \cdot w$
- Identity: $u \cdot 0 = 0 \cdot u = 0$ and $0 \in V$ for all $u \in V$
- Scalar multiplication: $u \cdot (-1) = -u$ and $u \cdot 1 = u$

Vector Spaces are very closely tied to the notion of "linear". "Linear" means respecting the structure of the vector space - vector addition, scalar multiplication (Linear Algebra is all about these two)

< vectors in $\mathbb{R}^n$ >

Vector V : $\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^n$

Geometrical Representation in 2D

Vector Addition: $\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} v_1 + w_1 \\ v_2 + w_2 \end{bmatrix}$

Scalar Multiplication: $\lambda \cdot \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \lambda v_1 \\ \lambda v_2 \end{bmatrix}$

Vector-Vector Product: $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$

Dot Product (Inner Product): $\vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i$

Outer Product: $\vec{a} \vec{b}^T \in \mathbb{R}^{n \times n}$

- The result of scalar multiplications across multiple vectors $\Rightarrow$ "Linear combination" $C_1 v_1 + C_2 v_2 + \dots + C_n v_n$
- The set of all possible Linear Combinations of the given vectors, is called the "Span" of that given vectors. $\text{span}(v_1, v_2) = \{ C_1 v_1 + C_2 v_2 \mid C_1, C_2 \in \mathbb{R} \}$
- A set of vectors $\{v_1, v_2, \dots, v_n\}$ is said to be "linearly independent" if $a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0 \Leftrightarrow a_1 = a_2 = \dots = a_n = 0$
- Proj(y ; $\{x_1, \dots, x_n\}$): Projection of a vector $y \in \mathbb{R}^n$ onto the span of $\{x_1, \dots, x_n\}$

Norm of a vector $\|v\|$

A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ that satisfies 4 properties

- for all $x \in \mathbb{R}^n$, $f(x) \geq 0$ (non-negativity)
- $f(x) = 0$ if and only if $x = 0$ (definiteness)
- for all $x, y \in \mathbb{R}^n$, $f(x+y) \leq f(x) + f(y)$ (triangle inequality)
- for all $x \in \mathbb{R}^n$, $f(\lambda x) = |\lambda| f(x)$ (homogeneity)

(Informally, a measure of the length of the vector)

Euclidean or L_2 norm: $\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$

L_1 norm: $\|x\|_1 = \sum_{i=1}^n |x_i|$

L_∞ norm: $\|x\|_\infty = \max_i |x_i|$

< Decoding vector spaces $\mathbb{R}^n$ >

Let $V \subset \mathbb{R}^n$ be a subset. We say that V is a subspace of $\mathbb{R}^n$ if:

- $0 \in V$
- $\vec{u}, \vec{v} \in V \Rightarrow \vec{u} + \vec{v} \in V$ (closed under sums)
- $\vec{v} \in V, \lambda \in \mathbb{R} \Rightarrow \lambda \vec{v} \in V$ (closed under scalar mult.)

Remark: Span is a subspace

For $\vec{u}_1, \dots, \vec{u}_n \in \mathbb{R}^n$: $\text{Span}\{\vec{u}_1, \dots, \vec{u}_n\} \subset \mathbb{R}^n$ is a subspace.

We want to show $\text{Span}\{\vec{u}_1, \vec{u}_2\}$ satisfies the three properties

- $0 \in \text{Span}\{\vec{u}_1, \vec{u}_2\}$ since $0 = 0 \cdot \vec{u}_1 + 0 \cdot \vec{u}_2$
- Take two vectors in span: $(a_1 \vec{u}_1 + a_2 \vec{u}_2) + (b_1 \vec{u}_1 + b_2 \vec{u}_2) = (a_1 + b_1) \vec{u}_1 + (a_2 + b_2) \vec{u}_2$
- $\lambda (a_1 \vec{u}_1 + a_2 \vec{u}_2) = (\lambda a_1) \vec{u}_1 + (\lambda a_2) \vec{u}_2$

Remark: The only finite vector space is $\{0\}$.

Let $V \subset \mathbb{R}^n$ be any subspace. Suppose $|V| = \text{number of vectors in } V < +\infty$, then $V = \{0\}$ (Finite)

Let V contain any non-zero vector $\vec{v}$, then $n \vec{v} \in V$ for all $n \in \mathbb{R}$. So a finite vector space only contains $\{0\}$.

Remark: Classification of spaces in low dimension

- $\mathbb{R}^1: \{0\}, \mathbb{R}$
- $\mathbb{R}^2: \{0\}, \text{Lines through origin}, \mathbb{R}^2$
- $\mathbb{R}^3: \{0\}, \text{Lines through origin}, \text{Planes through origin}, \mathbb{R}^3$

Proper Subspaces of $\mathbb{R}^n$ are exactly the flat objects through the origin

Let $V \subset \mathbb{R}^n$ be a vector space and $\{\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n\}$ be a basis for V .

We call $\dim V$ the number of vectors in $\mathcal{B}$.

Theorem - Uniqueness of dimension

Let $\mathcal{B}_1, \mathcal{B}_2$ be two bases of V .

Then $\# \text{vectors of } \mathcal{B}_1 = \# \text{vectors of } \mathcal{B}_2$.

$\Rightarrow$ This tells us dimension is well defined, it doesn't depend on which basis we choose, so we can speak of the dimension of V without ambiguity.

Remark: Coordinates with respect to a basis

$\vec{v} \in V$ (vector space)

Standard basis $\mathcal{E} = \{\vec{e}_1, \vec{e}_2, \dots, \vec{e}_n\}$

Basis $\mathcal{B} = \{\vec{b}_1, \vec{b}_2, \dots, \vec{b}_n\}$

$\vec{v} = \sum_{i=1}^n c_i \vec{e}_i = \sum_{j=1}^n d_j \vec{b}_j$

$\vec{v}$ represents the coordinates of $\vec{v}$ with respect to the basis $\mathcal{B}$.

$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$ such that $v_1 \vec{e}_1 + v_2 \vec{e}_2 + \dots + v_n \vec{e}_n = \vec{v}$

$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$ such that $v_1 \vec{b}_1 + v_2 \vec{b}_2 + \dots + v_n \vec{b}_n = \vec{v}$

Old view: $\vec{v} = (x, y)$

New view: $\vec{v}$ is abstract object (x, y) is just the representation

one way to write a vector: numbers are obsolete

many ways, one per basis: numbers depend on coord system

$\Rightarrow$ Big idea: A vector $\vec{v}$ itself is "the thing", but its numerical representation depends on which basis we use. Different basis $\Rightarrow$ different coordinates for the same vector

< Transformations of vector spaces - the linear ones >

For $\mathbb{R}^m \xrightarrow{T} \mathbb{R}^n$ functions, we can tell if the function is "linear".

(Preserves the structure of the source vector space. Intuitively, $\vec{0}$ stays at $\vec{0}$, grid lines stay parallel & evenly spaced)

Specifically, $T(\vec{0}) = \vec{0}$, $T(\vec{v}) + T(\vec{w}) = T(\vec{v} + \vec{w})$, $T(\lambda \vec{v}) = \lambda T(\vec{v})$

Remark: Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$: $S: \mathbb{R}^m \rightarrow \mathbb{R}^k$, and we have $(S \circ T): \mathbb{R}^n \rightarrow \mathbb{R}^k$, $\vec{v} \mapsto S(T(\vec{v}))$

- T, S are linear transformations $\Rightarrow S \circ T$ is a linear transformation
- $S \circ T$ is a linear transformation $\nRightarrow S$ or T is linear

ex) $T: \mathbb{R} \rightarrow \mathbb{R}$, $T(x) = x + 1$
 $S: \mathbb{R} \rightarrow \mathbb{R}$, $S(x) = x - 1$
 $(S \circ T)(x) = x$ / S or T is the zero function

Remark: For a set of independent vectors, the lin transformed output is still independent.

This "transformation" can be written in a matrix, by plugging in $\mathbb{R}^m$'s basis vectors into T and compiling them into a matrix, in order.

<Matrix definitions>

Matrix: Rectangular Arrays of numbers organized in rows and columns, used to solve linear equations, represent data, and define linear transformations.

matrix $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & & \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & \dots & \dots & a_{mn} \end{bmatrix} = \begin{bmatrix} | & & | \\ a^1 & \dots & a^n \\ | & & | \end{bmatrix} = \begin{bmatrix} -a_1^T- \\ \vdots \\ -a_n^T- \end{bmatrix} \in \mathbb{R}^{m \times n}$

<Special Matrices>

1) Zero Matrix

$$0 = \begin{bmatrix} 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix} \in M_n(\mathbb{R})$$

properties:

$$-A+0=A=0+A$$

$$-A \cdot 0=0=0 \cdot A$$

2) Identity Matrix

$$I = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix} \in M_{n,n}(\mathbb{R})$$

properties

$$-A \cdot I = I \cdot A = A$$

3) Diagonal Matrix

Matrix where all non-diagonal elements are 0

$$D = \text{diag}(d_1, d_2, \dots, d_n) \Rightarrow D_{ij} = \begin{cases} d_i & (i=j) \\ 0 & (i \neq j) \end{cases}$$

4) Transpose Matrix

def Given A, we can define A^T by flipping the numbers along the diagonal (rows $\leftrightarrow$ columns)

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \Rightarrow A^T = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$$

Theorem

$$(A+B)^T = A^T + B^T$$

$$(sA)^T = sA^T$$

$$(AC)^T = C^T A^T$$

Remark

$A = A^T$, A is called symmetric.

AA^T is always symmetric

<Matrix Properties>

def Norm of a Matrix

$$\text{Frobenius norm: } \|A\|_F = \sqrt{\sum_{i=1}^n \sum_{j=1}^n A_{ij}^2} = \sqrt{\text{tr}(A^T A)}$$

def

Trace of a square matrix

$$\text{tr} A = \sum_{i=1}^n A_{ii} \quad (A \in \mathbb{R}^{n \times n})$$

Theorem

$$\text{tr} A = \text{tr} A^T$$

$$\text{tr}(A+B) = \text{tr} A + \text{tr} B$$

$$\text{tr}(tA) = t \cdot \text{tr}(A) \quad (t \in \mathbb{R})$$

$$\text{if } AB \text{ is square, } \text{tr} AB = \text{tr} BA$$

$$\text{if } ABC \text{ is square, } \text{tr} ABC = \text{tr} BCA = \text{tr} CAB$$

5) Inverse Matrix

Finding Inverse Matrices

For $A \in M_n(\mathbb{R})$

1. Augment $(A|I)$

2. Row-reduce $(A|I) \rightarrow (I|B)$

3. Read off the inverse

$\hookrightarrow$ If A's n (pivot) = n (rows) / n (columns).

then A^{-1} and the right side B is the inverse A^{-1}

A^{-1} exists: A is invertible (non-singular)

Theorem

$$(AB)^{-1} = B^{-1} A^{-1}$$

$$(A^{-1})^{-1} = (A^{-1})^T$$

Remark

XX^T invertible $\Leftrightarrow$ X's columns lin. independent

XX^T invertible $\Leftrightarrow$ X's rows lin. independent

Inverse preserves definiteness.

6) Matrices with orthonormal columns

A's column vectors are orthonormal $\Leftrightarrow A^T A = I$

(col. vectors are orthogonal to one another and each has a norm of 1)

$$\|Ax\|_2 = \|x\|_2 \text{ for any matrix A w/ } A^T A = I \text{ and vector } x$$

When A is square, A is called an orthogonal matrix (U)

$$U^T U = I = U U^T \quad (U^{-1} = U^T)$$

added

Det can be +1 or -1

7) Rotational Matrix (R)

Orthogonal matrix with det 1

$$\text{def: } \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

$$R_{\text{rot}} = R A R^{-1} = R A R^T$$

<Basic Matrix Operations>

def

$$\text{addition: } A+B = \begin{bmatrix} (a_{11}+b_{11}) & \dots & (a_{1n}+b_{1n}) \\ \vdots & \ddots & \vdots \\ (a_{m1}+b_{m1}) & \dots & (a_{mn}+b_{mn}) \end{bmatrix}$$

def

$$\text{scalar multiplication: } cA = \begin{bmatrix} ca_{11} & \dots & ca_{1n} \\ \vdots & \ddots & \vdots \\ ca_{m1} & \dots & ca_{mn} \end{bmatrix}$$

Theorem

Let s, t: scalars, $A, B, C \in M_{n,m}(\mathbb{R})$

$$(a) A+B = B+A$$

$$(b) s(A+B) = sA + sB$$

$$(c) (s+t)A = sA + tA$$

$$(d) (A+B) + C = A + (B+C)$$

$$(e) (st)A = s(tA)$$

$$(f) A + 0_m = A$$

def

Matrix-Vector Product

we write A by rows

$$y = Ax = \begin{bmatrix} -a_1^T- \\ \vdots \\ -a_m^T- \end{bmatrix} x = \begin{bmatrix} a_1^T x \\ \vdots \\ a_m^T x \end{bmatrix}$$

$$y^T = x^T A = [x_1, x_2, \dots, x_m] \begin{bmatrix} -a_1^T- \\ \vdots \\ -a_m^T- \end{bmatrix}$$

$$= x_1 [-a_1^T] + \dots + x_m [-a_m^T]$$

we write A by columns

$$y = Ax = \begin{bmatrix} | & \dots & | \\ a^1 & \dots & a^n \\ | & \dots & | \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_n \end{bmatrix} = \begin{bmatrix} | & | & | \\ a_1^T x_1 + a_2^T x_2 + \dots + a^n^T x_n \\ | & | & | \end{bmatrix}$$

$$y^T = x^T A = x^T \begin{bmatrix} | & \dots & | \\ a^1 & \dots & a^n \\ | & \dots & | \end{bmatrix} = [x^T a^1 \quad x^T a^2 \quad \dots \quad x^T a^n]$$

Theorem

Let s be a scalar, and A, B, C matrices.

Then each of the following holds in the cases

where the indicated operations are defined

$$(a) A(BC) = (AB)C$$

$$(b) A(B+C) = AB+AC$$

$$(c) (A+B)C = AC+BC$$

$$(d) s(AB) = (sA)B = A(sB)$$

$$(e) AI = A \quad (I: \text{identity matrix of appropriate dimension})$$

$$(f) IA = A$$

def

Matrix-Matrix Product

As a set of dot products

$$C = AB = \begin{bmatrix} -a_1^T- \\ \vdots \\ -a_m^T- \end{bmatrix} \begin{bmatrix} | & \dots & | \\ b^1 & \dots & b^p \\ | & \dots & | \end{bmatrix} = \begin{bmatrix} a_1^T b^1 & \dots & a_1^T b^p \\ \vdots & \ddots & \vdots \\ a_m^T b^1 & \dots & a_m^T b^p \end{bmatrix}$$

As a sum of outer products

$$C = AB = \begin{bmatrix} | & \dots & | \\ a^1 & \dots & a^p \\ | & \dots & | \end{bmatrix} \begin{bmatrix} -b^1^T- \\ \vdots \\ -b^p^T- \end{bmatrix} = \sum_{i=1}^p a^i b^i{}^T$$

As a set of matrix-vector products

$$C = AB = A \begin{bmatrix} | & | & | & | \\ b^1 & b^2 & \dots & b^n \\ | & | & | & | \end{bmatrix} = \begin{bmatrix} | & | & | & | \\ Ab^1 & Ab^2 & \dots & Ab^n \\ | & | & | & | \end{bmatrix}$$

As a set of vector-matrix products

$$C = AB = \begin{bmatrix} -a_1^T- \\ \vdots \\ -a_m^T- \end{bmatrix} B = \begin{bmatrix} -a_1^T B- \\ \vdots \\ -a_m^T B- \end{bmatrix}$$

Theorem

Let A, B, C nonzero matrices

(a) If is possible that $AB \neq BA$

(b) $AB=0$ doesn't imply that $A=0$ or $B=0$

(c) $AC=BC$ doesn't imply that

$$A=B \text{ or } C=0$$

* Computing the Determinant

Determinant: A function $M_{n,n}(\mathbb{R}) \rightarrow \mathbb{R}$ such that $\det A \neq 0 \Leftrightarrow A$ is invertible.
 $A \mapsto \det A$ ($|A|$)

$n=1$ $\det: \mathbb{R} \rightarrow \mathbb{R}$ (since A isn't invertible only when $A=0$)
 $A \mapsto A$
 $A^{-1} = \frac{1}{A}$

$n=2$ $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$

A is invertible $\Leftrightarrow$ col vectors being lin. dependent

$\Leftrightarrow \exists \lambda \in \mathbb{R}: \vec{v}_1 = \lambda \vec{v}_2 \Leftrightarrow a_{11} = \lambda a_{21}, a_{21} = \lambda a_{22}, \frac{a_{11}}{a_{21}} = \frac{a_{21}}{a_{22}}$

$\Leftrightarrow a_{11}a_{22} - a_{21}a_{12} = 0$

$\Rightarrow$ we set $\det(A) = a_{11}a_{22} - a_{21}a_{12}$, then $\det(A) \neq 0 \Leftrightarrow A$ is invertible

$n=3$ (shortcut algorithm)

$$\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{matrix} a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} \end{matrix}$$

General definition with induction

First of all, for $A \in M_{n,n}(\mathbb{R})$ and $1 \leq i, j \leq n$,

we define A_{ij} = the matrix A without the i -th row and j -th column
 $= (i, j)$ -cofactor of $A \in M_{n-1, n-1}(\mathbb{R})$.

$\downarrow$
 Then we define the determinant inductively:

$n=1$: $\det: \mathbb{R} \rightarrow \mathbb{R}$
 $A \mapsto A$

$n-1 \rightarrow n$: assume $\det$ is defined for $(n-1) \times (n-1)$ matrices
 choose a row i (or row j)

then $\det(A) = a_{i1}a_{11} + a_{i2}a_{21} + \dots + a_{in}a_{in}$

where a_{ij} is the i th row j th column entry

and $a_{ij} = (-1)^{i+j} \cdot \det(A_{ij}) \in \mathbb{R}$

$\oplus$ For upper triangular matrices

$A = \begin{pmatrix} a_{11} & & * \\ & a_{22} & \\ 0 & & \ddots \\ & & & a_{nn} \end{pmatrix} \in M_{n,n}(\mathbb{R})$

$$\begin{aligned} \det(A) &= a_{nn}a_{nn} = a_{nn} \cdot (-1)^{n,n} \cdot \det \begin{pmatrix} a_{11} & & * \\ & \ddots & \\ 0 & & a_{n-1,n-1} \end{pmatrix} \\ &= a_{nn} \cdot a_{n-1,n-1} \cdot \det \begin{pmatrix} a_{11} & & * \\ & \ddots & \\ 0 & & a_{n-2,n-2} \end{pmatrix} \\ &= a_{nn} \cdot a_{n-1,n-1} \cdot \dots \cdot a_{11} \\ &\quad (\text{product of diagonal elements}) \end{aligned}$$

Remark

upper triangular matrix is invertible $\Leftrightarrow$ it has the maximum number of pivots

$\det(\text{upper triangular matrix}) \neq 0 \Leftrightarrow$ every diagonal entry is non-zero



Let A be a square matrix and B is an echelon form of A , then

$\det A = c \cdot (-1)^{\sigma} \det B$ (σ : # of times rows were switched, $c \in \mathbb{R}$)

A is invertible $\Leftrightarrow B$ is invertible $\Leftrightarrow \det B \neq 0 \Leftrightarrow \det A \neq 0$
 (Since B is upper-triangular)

The properties are:

0) $\det(\text{Id}) = 1$

1) determinant is linear on "single" rows

$\det \begin{pmatrix} \lambda \vec{v} + \mu \vec{w} \\ \vdots \\ \vec{v} \end{pmatrix} = \lambda \det \begin{pmatrix} \vec{v} \\ \vdots \\ \vec{v} \end{pmatrix} + \mu \det \begin{pmatrix} \vec{w} \\ \vdots \\ \vec{v} \end{pmatrix}$

Remark $\det(A+B) \neq \det(A) + \det(B)$

2) exchanging rows flips plus/minus

$\det \begin{pmatrix} \vec{v}_1 \\ \vec{v}_2 \\ \vdots \end{pmatrix} = - \det \begin{pmatrix} \vec{v}_2 \\ \vec{v}_1 \\ \vdots \end{pmatrix}$

Remark A matrix with two identical rows has a determinant of 0.
 ($\det A = -\det A$)

3) $\det(AB) = \det A \cdot \det B = \det B \cdot \det A = \det(BA)$ (Even though $AB \neq BA$)

$\det(A^{-1}) = \det(A)^{-1}$ (Since $\det(AA^{-1}) = 1 = \det(A) \det(A^{-1})$)

4) Inverses by determinant

Let $A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & \dots & \dots & a_{nn} \end{pmatrix} \Rightarrow \text{Co}(A) = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & \dots & \dots & a_{nn} \end{pmatrix}$
 $a_{ij} = (-1)^{i+j} \cdot \det(A_{ij})$

$A \cdot \text{Co}(A)^T = \det(A) \cdot \text{Id}$

$A^{-1} = \frac{1}{\det(A)} (\text{Co}(A))^T$

Remark

$\det \begin{pmatrix} I \\ T \\ B \end{pmatrix} = \det(B^{-1} [T]_E^E B) = \det(B)^{-1} \cdot \det[T]_E^E \cdot \det(B)$
 $= \det([T]_E^E)$

Theorem
 $\Rightarrow \mathcal{A}: M_{n,n}(\mathbb{R}) \rightarrow \mathbb{R}$ verifies
 $(a_1, \dots, a_n)$, then $\mathcal{A} = \det$.

* Eigen-Theory (Square matrices only)

Some nonzero vectors stay in the same direction during transformations.

$$A\vec{v} = \lambda\vec{v} \quad V_\lambda = \{\vec{v} \in \mathbb{R}^n \mid A\vec{v} = \lambda\vec{v}\} \quad (\text{set of all eigenvectors } + \vec{0})$$

eigenvector eigenvalue eigenspace

Remark

• For diff. eigenvalues, their eigenvectors are lin. independent

§

Then we know: $A\vec{v} = \lambda\vec{v} \Leftrightarrow (A - \lambda I)\vec{v} = \vec{0}$

§

$$V_\lambda = \ker(A - \lambda I)$$

Then we know:

$$\lambda \text{ is an eigenvalue} \Leftrightarrow \dim V_\lambda \geq 1 \Leftrightarrow \dim \ker(A - \lambda I) \geq 1$$

↗

$$\det(A - \lambda I) = 0 \Leftrightarrow A - \lambda I \text{ matrix not invertible} \Leftrightarrow A - \lambda I \text{ matrix not injective}$$

polynomial in λ of degree n

"characteristic polynomial of A "

Remark

Number of roots (eigenvalues) = number of nonzero eigenspaces
at most n

Remark

$$\det([A - \lambda I]_{\mathcal{B}}^{\mathcal{B}}) = \det([A - \lambda I]_{\mathcal{E}}^{\mathcal{E}})$$

↳ $P_A(\lambda)$ is well-defined (doesn't depend on the choice of a basis)

Theorem

- $\text{tr} A = \sum_{i=1}^n \lambda_i$
- $\det A = \prod_{i=1}^n \lambda_i$
- $\text{rk}(A) = \text{number of nonzero eigenvalues}$
- Suppose A is invertible (non-singular) with eigenvalue λ and associated eigenvector η .
Then $A^{-1}\eta = \left(\frac{1}{\lambda}\right)\eta$
- Eigenvalues of a diagonal matrix are just the diagonal entries.
- For a matrix to be invertible, all eigenvalues must be non-zero.

Diagonalization (Eigen - Decomposition)

Out of the various possibilities we can write transformation matrices, which is the simplest?

→ What if we only collected lin. independent eigenvectors and declared them as basis?

Let B the basis consisted of only eigenvectors.

$$[A]_B^B = B^{-1} [A]_E^E B$$

$$= \left(\begin{array}{c|c|c|c} | & | & | & | \\ [T(B_1)]_B & [T(B_2)]_B & \dots & \dots \\ | & | & | & | \end{array} \right) = \left(\begin{array}{c|c|c|c} | & | & | & | \\ [A_1 \vec{b}_1]_B & [A_2 \vec{b}_2]_B & \dots & \dots \\ | & | & | & | \end{array} \right) = \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{pmatrix} =: D$$

$$A = B D B^{-1} \rightarrow \text{This is called "diagonalizing"}$$

Remark

- In order for A to be diagonalizable, there has to be n lin. independent eigenvectors.
- A, B, D, B⁻¹ are all same sizes

However what if we can't diagonalize:

1) There were no (real number) eigenvalues at all

2) For matrices like $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, char. polynomial = $\det \begin{pmatrix} 1-\lambda & 1 \\ 0 & 1-\lambda \end{pmatrix} = (1-\lambda)^2$, $\dim \ker \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = 1$



To help judgement we came up with the following:

We call $\left\{ \begin{array}{l} \text{the multiplicity of } \lambda \text{ in char. pol "algebraic multiplicity"} - m_a(\lambda) \\ \text{Remark} \\ m_a(\lambda_1) + \dots + m_a(\lambda_r) \leq n \text{ (= if complex numbers)} \end{array} \right.$

$\left\{ \begin{array}{l} \text{the number of lin. independent eigenvectors} \\ \text{of eigenvalue } \lambda \text{ "geometric multiplicity"} - m_g(\lambda) \end{array} \right.$

Remark
This is $\dim V_\lambda$

and $1 \leq m_g(\lambda) \leq m_a(\lambda) \leq n$

eigenvectors eigenvalues

$$m_g(\lambda_1)=3 \left(\begin{array}{l} \text{span}(V_1) \\ \text{span}(V_2) \\ \text{span}(V_3) \end{array} \right) = \lambda_1$$

$$m_g(\lambda_2)=2 \left(\begin{array}{l} \text{span}(V_4) \\ \text{span}(V_5) \end{array} \right) = \lambda_2$$

$$m_g(\lambda_r)=1 \left(\text{span}(V_k) \right) = \lambda_r$$

* $V_1, \dots, V_k$ are lin. independent

* A is diagonalizable $\Leftrightarrow$ n lin. independent eigenvectors exist $\Leftrightarrow m_g(\lambda_1) + \dots + m_g(\lambda_r) = n \Leftrightarrow \begin{array}{l} m_a(\lambda_1) + \dots + m_a(\lambda_r) = n \\ \&\& \\ m_a(\lambda) = m_g(\lambda) \end{array}$

If A is diagonalizable, we can extract n lin. independent eigenvectors and make them into a matrix:

$$\left(\begin{array}{c|c|c|c|c|c|c|c} | & | & | & | & | & | & | & | \\ V_{\lambda_1} & V_{\lambda_1} & V_{\lambda_2} & V_{\lambda_2} & V_{\lambda_2} & V_{\lambda_2} & \dots & V_{\lambda_r} \\ | & | & | & | & | & | & | & | \end{array} \right) = B, \text{ then } \begin{pmatrix} \lambda_1 & & & & \\ & \lambda_1 & & & \\ & & \lambda_2 & & \\ & & & \ddots & \\ 0 & & & & \lambda_r \end{pmatrix} = D \text{ (Order matters)}$$

Remark

$$\det(A) = \det(D) = \lambda_1 \times \lambda_2 \times \dots \times \lambda_n$$

Remark

If λ = eigenvalue & $\vec{v}$ = eigenvector, then for an arbitrary polynomial q: $q(A)\vec{v} = q(\lambda)\vec{v}$

Theorem * Spectral Theorem

For symmetric matrices,

- all eigenvalues are real numbers
- it is diagonalizable, and B is orthogonal
(n lin. independent eigenvectors)

$$A = U \Lambda U^{-1} = U \Lambda U^T$$

* The quadratic form & Definiteness

$A \in \mathbb{R}^{n \times n}$, $x \in \mathbb{R}^n$, the scalar value $x^T A x$ is called a quadratic form.

$$x^T A x = \sum_{i=1}^n x_i (Ax)_i = \sum_{i=1}^n x_i \left(\sum_{j=1}^n A_{ij} x_j \right) = \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j$$

Remark

Since $x^T A x$ is a scalar, $x^T A x = (x^T A x)^T = x^T A^T x = x^T \left(\frac{1}{2} A + \frac{1}{2} A^T \right) x$

$\frac{1}{2} A + \frac{1}{2} A^T$ is always symmetric, so we implicitly assume that the matrix appearing in a quadratic form is symmetric.

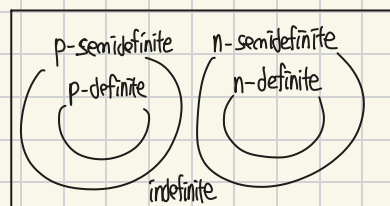
Remark

$$\begin{aligned} x^T A x &= x^T U \Lambda U^T x = (\tilde{x}^T U)^T \Lambda (U^T x) = (\tilde{x}^T \tilde{x})^T \Lambda (\tilde{x}) \\ &= \sum_{i=1}^n \lambda_i \tilde{x}_i^2 \quad (\tilde{x} = [x]_U) \end{aligned}$$

If $A = A^T$ and

$x^T A x > 0$ for all non-zero vectors, A is positive definite. $A \succ 0$
 ≥ 0 for all vectors, A is positive semidefinite. $A \succeq 0$
 < 0 for all non-zero vectors, A is negative definite. $A \prec 0$
 ≤ 0 for all vectors, A is negative semidefinite. $A \preceq 0$

If $x^T A x$ could both be positive and negative, A is indefinite.



Remark

The definiteness of the matrix A depends entirely on the sign of its eigenvalues.

If all $\lambda_i > 0 \rightarrow \sum_{i=1}^n \lambda_i \tilde{x}_i^2 > 0$ for any $\tilde{x} \neq 0 \rightarrow A$ is positive definite
 $\geq 0 \rightarrow \sum_{i=1}^n \lambda_i \tilde{x}_i^2 \geq 0$ for all $\tilde{x} \rightarrow A$ is positive semidefinite
 $< 0 \rightarrow \sum_{i=1}^n \lambda_i \tilde{x}_i^2 < 0$ for any $\tilde{x} \neq 0 \rightarrow A$ is negative definite
 $\leq 0 \rightarrow \sum_{i=1}^n \lambda_i \tilde{x}_i^2 \leq 0$ for all $\tilde{x} \rightarrow A$ is negative semidefinite
 both positive & negative eigenvalues $\rightarrow A$ is indefinite

Matrix Calculus

* The gradient

$f: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ (function that takes a $m \times n$ matrix A and returns a real value)

Then the gradient of f at A is the matrix of A 's partial derivatives

$$\nabla_A f(A) = \begin{bmatrix} \frac{\partial f(A)}{\partial A_{11}} & \dots & \frac{\partial f(A)}{\partial A_{1n}} \\ \vdots & \ddots & \vdots \\ \frac{\partial f(A)}{\partial A_{m1}} & \dots & \frac{\partial f(A)}{\partial A_{mn}} \end{bmatrix} \in \mathbb{R}^{m \times n}$$

i.e. $(\nabla_A f(A))_{ij} = \frac{\partial f(A)}{\partial A_{ij}}$

It follows directly from the equivalent properties of partial derivatives

$$\nabla_A (f(A) + g(A)) = \nabla_A f(A) + \nabla_A g(A)$$

for $t \in \mathbb{R}$, $\nabla_A (t f(A)) = t \nabla_A f(A)$

* The Hessian

$f: \mathbb{R}^n \rightarrow \mathbb{R}$

Then the Hessian matrix with respect to x is the $n \times n$ matrix of 2nd partial derivatives

$$\nabla_x^2 f(x) = \begin{bmatrix} \frac{\partial^2 f(x)}{\partial x_1^2} & \dots & \frac{\partial^2 f(x)}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f(x)}{\partial x_n \partial x_1} & \dots & \frac{\partial^2 f(x)}{\partial x_n^2} \end{bmatrix} \rightarrow \text{earlier}$$

i.e. $(\nabla_x^2 f(x))_{ij} = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}$

Remark Hessian is always symmetric, since $\frac{\partial^2 f(x)}{\partial x_i \partial x_j} = \frac{\partial^2 f(x)}{\partial x_j \partial x_i}$

* The Jacobian

For a vector-valued function $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$ containing m output equations and n input variables:

$$F(x_1, \dots, x_n) = \begin{bmatrix} F_1(x_1, \dots, x_n) \\ F_2(x_1, \dots, x_n) \\ \vdots \\ F_m(x_1, \dots, x_n) \end{bmatrix} \quad m$$

The Jacobian matrix J is an $m \times n$ matrix where i th row, j th column entry is the partial derivative of the i th function with respect to the j -th variable

$$J = \begin{bmatrix} \frac{\partial F_1}{\partial x_1} & \dots & \frac{\partial F_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_1} & \dots & \frac{\partial F_m}{\partial x_n} \end{bmatrix}$$

* Gradients of linear functions

For $x \in \mathbb{R}^n$, let $f(x) = b^T x$ for some known vector $b \in \mathbb{R}^n$

Then $f(x) = \sum_{i=1}^n b_i x_i$, so $\frac{\partial f(x)}{\partial x_k} = \frac{\partial}{\partial x_k} (b_1 x_1 + b_2 x_2 + \dots + b_n x_n) = b_k$

Generalise: $\nabla_x b^T x = b$ (analogous situation: $\frac{\partial}{\partial x} ax = a$) and of course $\nabla_x^2 b^T x = 0$

* Gradients of quadratic functions

$x \in \mathbb{R}^n$, $f(x) = x^T A x$ for $A \in \mathbb{S}^n$ (nonsymmetric matrix)

$$\frac{\partial f(x)}{\partial x_k} = \frac{\partial}{\partial x_k} \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j$$

$$= \frac{\partial}{\partial x_k} \left[\sum_{i \neq k} \sum_{j \neq k} A_{ij} x_i x_j + \sum_{i \neq k} \sum_{j=k} A_{ij} x_i x_j + \sum_{i=k} \sum_{j \neq k} A_{ij} x_i x_j + \sum_{i=k} \sum_{j=k} A_{ij} x_i x_j \right]$$

$$= \frac{\partial}{\partial x_k} \left[\sum_{i \neq k} A_{ik} x_i x_k + \sum_{j \neq k} A_{kj} x_k x_j + A_{kk} x_k^2 \right]$$

$$= \sum_{i \neq k} A_{ik} x_i + \sum_{j \neq k} A_{kj} x_j + 2 A_{kk} x_k$$

$$= \sum_{i=1}^n A_{ik} x_i + \sum_{j=1}^n A_{kj} x_j = 2 \sum_{i=1}^n A_{ki} x_i$$

Generalise: $\nabla_x x^T A x = 2 A x = (A + A^T) x$

True even when A isn't symmetric

• $\nabla_x b^T x = b$

• $\nabla_x x^T b = b$

• $\nabla_x^2 b^T x = 0$

• $\nabla_x x^T A x = (A + A^T) x$

• $\nabla_x^2 x^T A x = A + A^T$

* Hessians of quadratic functions

$$\frac{\partial^2 f(x)}{\partial x_k \partial x_l} = \frac{\partial}{\partial x_k} \left[\frac{\partial f(x)}{\partial x_l} \right] = \frac{\partial}{\partial x_k} \left[2 \sum_{i=1}^n A_{li} x_i \right] = 2 A_{lk} = 2 A_{kl}$$

• $\nabla_x^2 x^T A x = 2A$

Remark

The Hessian of a convex function is positive-semi-definite.

Symmetric Matrix A

- All eigenvalues are real numbers

- Diagonalizable, and P is orthogonal

- PD/PSD/ND/NSD/ID

- Matrix is PD/ND $\rightarrow$ full rank (invertible)

(not full rank $\rightarrow$ not PD/ND)

- PSD = symmetric + all eigenvalues ≥ 0

$\hookrightarrow X^T X$ (Gram matrix - symmetric)

$\rightarrow$ PSD

- X 's columns lin. independent $\rightarrow X^T X$ is invertible, PD (if it's $X X^T$, X 's rows have to be lin. independent)

* Hessian (Always symmetric) Original Function

PSD at every point $\leftrightarrow$ Convex

PD at every point $\rightarrow$ Strictly convex

CE: x^*

PCA (Principal Component Analysis)

$X \in \mathbb{R}^{n \times d}$ Assume X is normalized $\left(\mu_j = \frac{1}{n} \sum_{i=1}^n x_{ij}, \sigma_j = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_{ij} - \mu_j)^2} \right)$
 $a_{ij} \leftarrow \frac{x_{ij} - \mu_j}{\sigma_j}$

$\frac{1}{n} X^T X = V L V^T$
 Diagonalize

- Since $X^T X$ is symmetric, $V V^T = I$
- $L = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_d \end{pmatrix}$

Re-arrange V, L s.t. $\lambda_1 \geq \lambda_2 \geq \dots \lambda_d \geq 0$

pick a k ($< d$) that will be the reduced dimension

put together k eigenvectors that correspond to the k largest eigenvalues (vertically) into a matrix

$$\tilde{V} = \begin{pmatrix} & & k & & \\ | & & & & \\ | & & & & \\ | & & & & \\ & \dots\dots & & & \\ | & & & & \\ & & & & d \end{pmatrix}$$

(This can be thought of as the 'parameter' we're learning.)

This is the basis vector for our new subspace

Then we get $\tilde{X} = X\hat{V} \in \mathbb{R}^{n \times k}$

→ We're trying to move all data from $\mathbb{R}^d$ into a k -dimensional subspace. How?

* Interpretation 1: Data's variance should be maximized after moving
s.t. X is only moved with XA (i.e. $A^T X^T$) and $AA^T = I_k$

How do we quantify the variance of $\begin{pmatrix} \text{---} k \text{---} \\ XA \end{pmatrix}$? → Sum of k features' individual variances
= trace (Cov matrix $\frac{1}{n} A^T X^T X A \in \mathbb{R}^{k \times k}$)

$$\begin{pmatrix} \text{---} d \text{---} \\ X \end{pmatrix} \begin{pmatrix} \text{---} k \text{---} \\ A \end{pmatrix} = \begin{pmatrix} \text{---} k \text{---} \\ XA \end{pmatrix} \begin{pmatrix} \text{---} n \text{---} \end{pmatrix}$$

$A_1 \sim A_k$

$$\frac{1}{n} \cdot \begin{pmatrix} \text{---} n \text{---} \\ \begin{pmatrix} X^{(1)} A_1 & \dots & X^{(n)} A_1 \\ \vdots & \ddots & \vdots \\ X^{(1)} A_k & \dots & X^{(n)} A_k \end{pmatrix} \begin{pmatrix} X^{(1)} A_1 & \dots & X^{(n)} A_1 \\ \vdots & \ddots & \vdots \\ X^{(1)} A_k & \dots & X^{(n)} A_k \end{pmatrix} \end{pmatrix} \begin{pmatrix} \text{---} k \text{---} \end{pmatrix} = \begin{pmatrix} \frac{1}{n} \sum_{i=1}^n (X^{(i)} A_1)^2 & \dots & \frac{1}{n} \sum_{i=1}^n (X^{(i)} A_1 X^{(i)} A_k) \\ \vdots & \ddots & \vdots \\ \frac{1}{n} \sum_{i=1}^n (X^{(i)} A_k X^{(i)} A_1) & \dots & \frac{1}{n} \sum_{i=1}^n (X^{(i)} A_k)^2 \end{pmatrix} \begin{pmatrix} \text{---} k \text{---} \end{pmatrix}$$

so $\arg\max_A \text{trace}(\frac{1}{n} A^T X^T X A)$, s.t. $A^T A = I_k$. Denote $\frac{1}{n} X^T X$ as Σ

→ Construct Lagrangian $\mathcal{L}(A, B) = \text{tr}(A^T \Sigma A) - B \cdot \text{tr}(A^T A - I_k)$

$$\frac{\partial}{\partial A} \mathcal{L}(A, B) = 2 \Sigma A - 2 A B = 0 \Rightarrow \Sigma A = A B \Rightarrow \bar{A} \Sigma A = B$$

so $\arg\max_A (\dots)$ is when $A^T \Sigma A = B$, and problem becomes $\arg\max_A \text{tr}(B)$

What is B ?

$$\begin{pmatrix} \text{---} d \text{---} \\ \Sigma = \frac{1}{n} X^T X \end{pmatrix} \begin{pmatrix} \text{---} k \text{---} \\ A \end{pmatrix} = \begin{pmatrix} \text{---} k \text{---} \\ A \end{pmatrix} \begin{pmatrix} \text{---} k \text{---} \\ B \end{pmatrix}$$

Since at our solution $B = A^T \Sigma A$ is a symmetric matrix, it is diagonalizable.

Let $B = \begin{pmatrix} \lambda_{B1} & & 0 \\ & \ddots & \\ 0 & & \lambda_{Bk} \end{pmatrix}$, then $\Sigma A_j = \lambda_{Bj} A_j$, so $\lambda_{B1} \sim \lambda_{Bk}$ turn out to be Σ 's eigenvalues
and A 's columns turn out to be Σ 's eigenvectors.

So $B = \text{diag}(k \text{ eigenvalues picked from } \Sigma\text{'s } d \text{ eigenvalues})$ & $A = k \text{ corresponding eigenvectors of } \Sigma$

$\text{tr}(B) = \text{sum of whichever } k \text{ eigenvalues we picked, so pick the highest } k \text{ values}$

$\arg\max_A \text{tr}(B) = \Sigma\text{'s eigenvectors with the largest eigenvalues}$

⊕ How do we pick the right k ?

If $k=d$, we will capture all the variance = $\lambda_1 + \dots + \lambda_d$

$$\text{Variance captured} = \frac{\sum_{j=1}^k \lambda_j}{\sum_{j=1}^d \lambda_j}$$

We want to use the least k to capture as much variance as possible

So set a percentage criterion, and increase k until you get it

* Interpretation 2: The data points' movement should be at the minimum

$$\underset{\mathbb{R}^{n \times d}}{X} \xrightarrow{\cdot A} \underset{\mathbb{R}^{n \times k}}{XA} \xrightarrow{\cdot A^T} \underset{\mathbb{R}^{n \times d}}{XAA^T} = \hat{X}$$

Since $AA^T \neq I_d$, $X \neq \hat{X}$. so what's $\underset{A}{\operatorname{argmin}} \|X - \hat{X}\|_F^2$ s.t. $A^T A = I_k$?

First, note that $X - \hat{X} = X - XAA^T = R$, then $RA = XA - XAA^T A = XA - XA = 0$.

So R 's each row (each example's difference) is orthogonal to A 's every column vector.

$$\text{Then } \hat{X} R^T = XAA^T R^T = XA(RA)^T = 0.$$

So $\hat{X}$'s every row is orthogonal to R 's every row.

Then, for each example, $x = \hat{x} + r$, and $\hat{x} \perp r$

$$\Rightarrow \|x\|_2^2 = \|\hat{x}\|_2^2 + \|r\|_2^2, \|\hat{x}\|_2^2 = \|xAA^T\|_2^2 = xAA^T A A^T x^T = \|xA\|_2^2$$

$$\Rightarrow \|x\|_2^2 = \|xA\|_2^2 + \|r\|_2^2$$

$$\Rightarrow \sum_{i=1}^n \|x\|_2^2 = \sum_{i=1}^n \|x^{(i)} A\|_2^2 + \sum_{i=1}^n \|r^{(i)}\|_2^2$$

$$\Rightarrow \underbrace{\frac{1}{n} \sum_{i=1}^n \|x\|_2^2}_{\operatorname{tr}(\Sigma)} = \underbrace{\frac{1}{n} \sum_{i=1}^n \|x^{(i)} A\|_2^2}_{\operatorname{tr}(A^T \Sigma A)} + \underbrace{\frac{1}{n} \sum_{i=1}^n \|r^{(i)}\|_2^2}_{\text{Reconstruction Error (MSE)}}$$

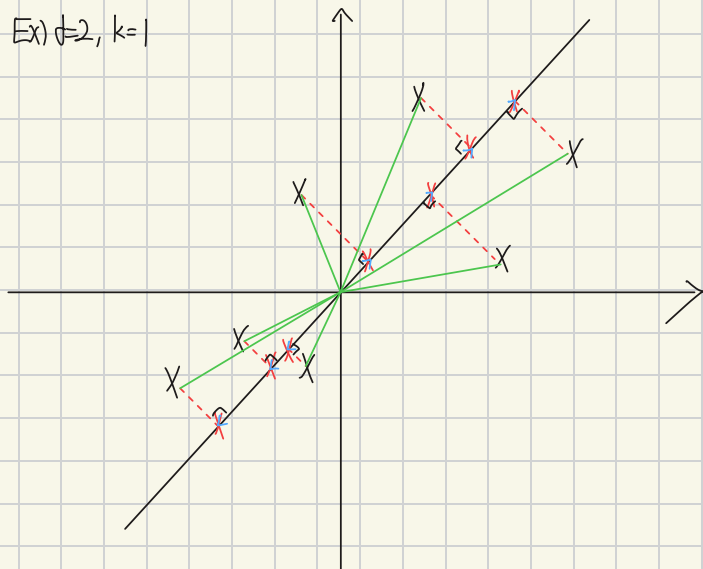
(The variance we threw away)

$$\Rightarrow \frac{1}{n} \|X\|_F^2 = \frac{1}{n} \|XA\|_F^2 + \frac{1}{n} \|X - \hat{X}\|_F^2$$

$$\text{So } \underset{A}{\operatorname{argmin}} \|X - \hat{X}\|_F^2 = \underset{A}{\operatorname{argmax}} (\operatorname{tr}(A^T \Sigma A))$$

$\therefore$ Same as Interpret. 1 (Maximizing Variance)

Ex) $d=2, k=1$



$$\underbrace{\frac{1}{n} \sum_{i=1}^n \|x\|_2^2}_{\operatorname{tr}(\Sigma)} = \underbrace{\frac{1}{n} \sum_{i=1}^n \|x^{(i)} A\|_2^2}_{\operatorname{tr}(A^T \Sigma A)} + \underbrace{\frac{1}{n} \sum_{i=1}^n \|r^{(i)}\|_2^2}_{\text{Reconstruction Error (MSE)}}$$

mean of all the (green line's length)² mean of all the (blue line's length)² mean of all the (red line's length)²

SVD (Singular Value Decomposition) for PCA

$$X \in \mathbb{R}^{n \times d} \quad (\text{Also assume } X \text{ is normalized})$$

↳ SVD (Doesn't need to compute $X^T X$)

SVD: A general matrix factorization technique
(Not a learning algorithm itself)

Here used as the numerically stable way to compute PCA

$$X = \underset{\tilde{n} \times \tilde{n}}{U} \underset{\tilde{n} \times k}{\Sigma} \underset{\tilde{k} \times d}{V^T}$$

$$\cdot U^T U = I$$

$$\cdot V^T V = I$$

$$\cdot \Sigma = \begin{pmatrix} \diagup & & \\ & \ddots & \\ 0 & & \diagdown \end{pmatrix} / \begin{pmatrix} \diagup & & \\ & 0 & \\ \vdots & & \ddots \\ & & & 0 \end{pmatrix}$$

• U : eigenvectors of XX^T

• V : eigenvectors of $X^T X$ (same to PCA's V)

$$\cdot \Sigma \Sigma^T = nI \quad (X^T X = V \Sigma^T U^T U \Sigma V^T = V \Sigma^T \Sigma V^T)$$

(Again, need to sort columns so singular values will be arranged in decreasing order (left → right))

↳

put together k eigenvectors that correspond to the k largest eigenvalues (vertically) into a matrix

$$\tilde{V} = \begin{pmatrix} | & | & \dots & | \\ | & | & \dots & | \\ | & | & \dots & | \\ | & | & \dots & | \end{pmatrix} \quad \begin{matrix} \text{--- } k \text{ ---} \\ \text{--- } d \text{ ---} \end{matrix}$$

This is the basis vector for our new subspace

↳

$$\hat{X} = X \tilde{V} = U \Sigma V^T \tilde{V} = U \hat{\Sigma} \quad (\hat{\Sigma} \text{ is the front } k \text{ columns of } \Sigma)$$

(Just hold on to U, Σ picked from above and pick k columns from Σ directly)

(So no need to actually construct $\tilde{V}$)

Fixes:

1. Condition number of a matrix: $\frac{\sigma_{\max}}{\sigma_{\min}}$

• Lower = more numerically stable

• $\sigma = |\lambda|$ for symmetric matrices

$$k(X) = \frac{\sigma_{\max}}{\sigma_{\min}}, \quad k(X^T X) = \frac{\lambda_{\max}(X^T X)}{\lambda_{\min}(X^T X)} = \frac{(\sigma_{\max}(X))^2}{(\sigma_{\min}(X))^2} = (k(X))^2$$

SVD: No need to even construct $X^T X$

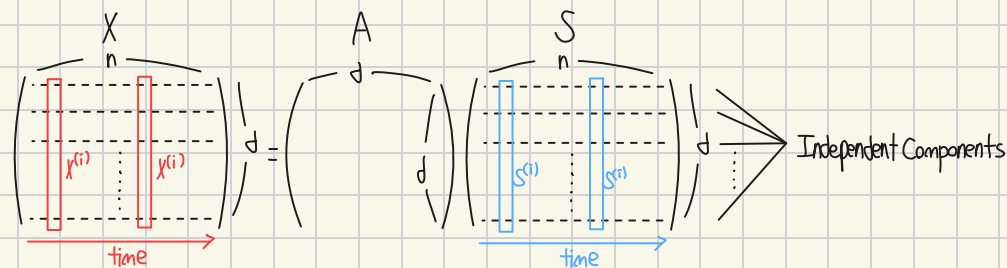
2. No need to diagonalize $X^T X$ (expensive computation when d gets big)

ICA (Independent Component Analysis)

$$x^{(i)} = A s^{(i)} \quad * i = \text{time}$$

data we see mixing matrix source

Problem: What is $A^{-1} = W$?



(ICA Ambiguities)

- Scale: $x = As = (\frac{A}{c})(cs)$
 $\Rightarrow$ solution: normalize s
- Permutation: $x = As = (AP^{-1})(P_s)$ // $S_1 \leftrightarrow S_2, A_1 \leftrightarrow A_2$: same result
 $\Rightarrow$ no problem since it's just the order of the independent components changing

- Orthogonality & Gaussian data: $x \sim N(0, AA^T)$
 $x' = ARS$ ($R = \text{orthogonal matrix}, R^T R = R R^T = I$)
 $= A's$
 $x' \sim N(0, A'A'^T)$
 $\sim N(0, AA^T)$

$\Rightarrow$ Can't tell difference between many (A, s) possibilities $\Rightarrow$ ICA requires S to be non-Gaussian
 (At most one component may be gaussian)

Assume S_j follows non-Gaussian distribution $p_s(s_j)$

$\hookrightarrow$ Set CDF as a sigmoid: $g_s(s_j) = \frac{1}{1+e^{-s_j}}$ $p_s(s_j) = g'_s(s_j)$ (Any non-gaussian CDF will work)

$$p(s) = \prod_{j=1}^d p_s(s_j) \quad // \text{ Since we're assuming the sources are independent}$$

$$\text{since } s = Wx, \quad p(x) = \prod_{j=1}^d p_s(W_j^T x) \cdot |\det(W)|$$

$$\text{So } \ell(w) = \sum_{i=1}^n \left(\sum_{j=1}^d [\log g'_s(w_j^T x^{(i)})] + \log |\det(W)| \right)$$

$\Rightarrow$ Randomly init W , do SGD

$$W := W + \alpha \frac{\partial \ell(w)}{\partial W}$$

$$:= W + \alpha \begin{pmatrix} 1 - 2g(w_1^T x^{(i)}) \\ \vdots \\ 1 - 2g(w_d^T x^{(i)}) \end{pmatrix} x^{(i)T} + (W^T)^{-1}$$